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Report
ON
Autism Spectrum Disorder Prediction

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Autism Spectrum Disorder Prediction

1. Introduction

Autism Spectrum Disorder (ASD) is a neurological condition that affects communication, behavior, and social interaction. Early detection is very important for timely treatment and support. Traditional diagnosis methods are slow and require expert analysis. This project uses machine learning to predict ASD quickly and efficiently using screening data.

2. Data Description

The dataset is taken from Kaggle and contains patient screening data.

Features include:

Age, Gender, Ethnicity, Family history, AQ-10 screening scores

Target variable: ASD (Yes/No)

EDA Findings:

- Dataset cleaned and preprocessed
- Categorical data encoded
- AQ score features strongly related to ASD
- Minimal missing values after handling

	age	gender	ethnicity	jundice	austim	contry_of_res	used_app_before	relation
0	26	f	White-European	no	no	'United States'	no	Self
1	24	m	Latino	no	yes	Brazil	no	Self
2	27	m	Latino	yes	yes	Spain	no	Parent
3	35	f	White-European	no	yes	'United States'	no	Self
4	40	f	?	no	no	Egypt	no	?
...	...	...	...	...	...	...	...	...
699	25	f	White-European	no	no	Russia	no	Self
700	34	m	Hispanic	no	no	Mexico	no	Parent
701	24	f	?	no	no	Russia	no	?
702	35	m	'South Asian'	no	no	Pakistan	no	Self
703	26	f	White-European	no	no	Cyprus	no	Self

704 rows × 8 columns

Figure 1: Class Distribution of ASD and Non-ASD Cases

Figure 1 represents the distribution of target classes in the dataset. It helps in understanding whether the dataset is balanced or imbalanced. Class distribution is important because highly imbalanced data can affect model performance and prediction accuracy. This analysis provides insight into the dataset before model training.

3. Feature Importance

Using SHAP analysis:

- AQ scores are most important features
- Family history also impacts prediction
- Demographic features have less impact

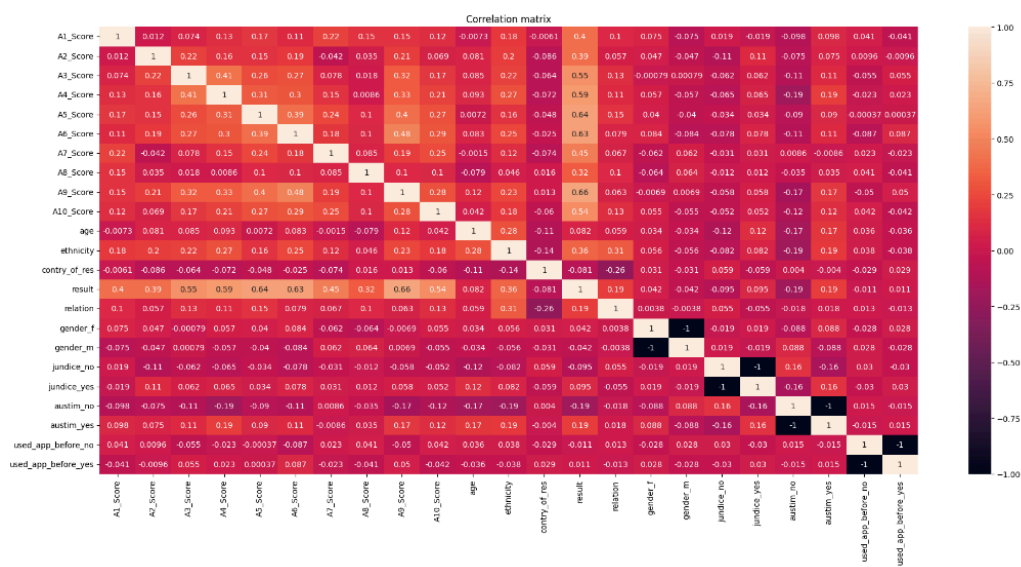


Figure 2: SHAP-Based Feature Importance for ASD Prediction

Figure 2 illustrates the contribution of each feature to the prediction made by the machine learning model. SHAP analysis shows that AQ-10 screening scores are the most influential features in identifying ASD. Family history also contributes significantly, while demographic features have relatively less impact. This improves the interpretability of the prediction model.

4. Methodology

Flow of project:

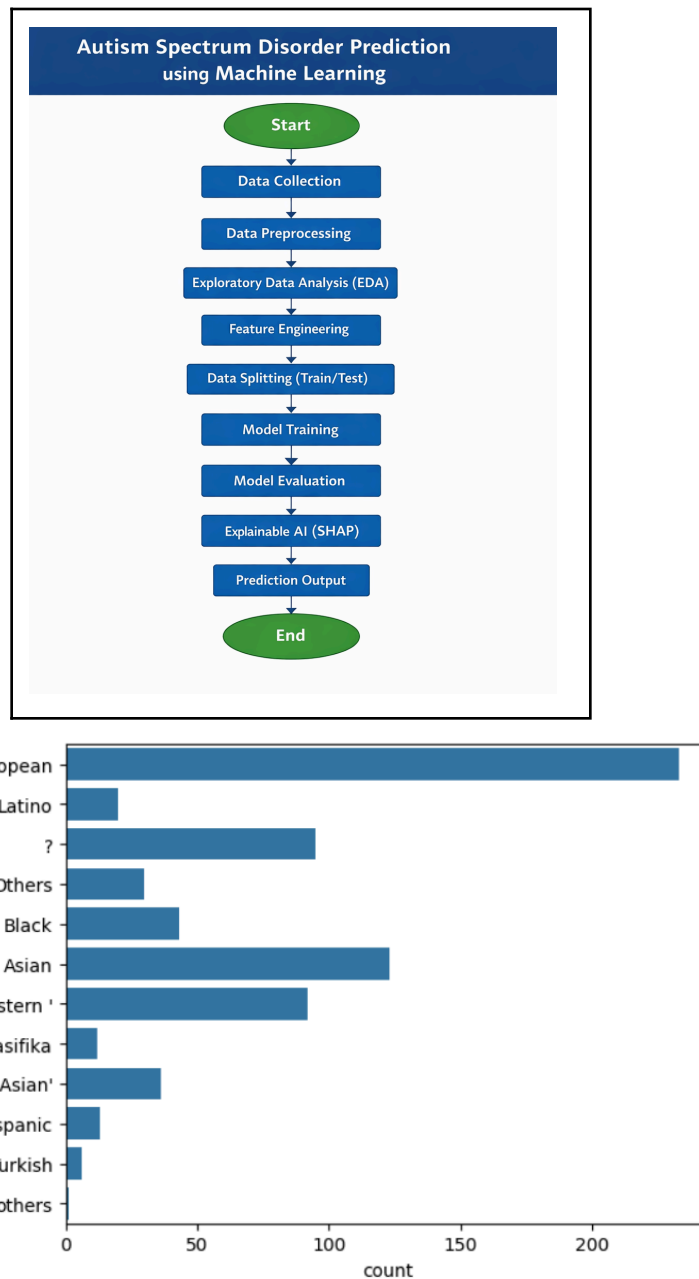


Figure 3: Workflow of Autism Spectrum Disorder Prediction System

Figure 3 shows the complete workflow followed in this project. The process starts with data collection from the Kaggle ASD dataset, followed by preprocessing, encoding, and exploratory data analysis. After feature selection, multiple machine learning models were trained and evaluated using

standard performance metrics. The final step involves comparing the models to identify the best algorithm for ASD prediction.

5. Machine Learning Algorithms

- Logistic Regression – simple and interpretable
- KNN – based on nearest neighbors
- Naïve Bayes – probability-based
- Decision Tree – tree structure model
- SVM – finds optimal boundary
- Random Forest – ensemble of trees
- Gradient Boosting – sequential learning
- XGBoost – optimized boosting algorithm

6. Result and Discussion

Evaluation Metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix

Evaluation Parameters:

Accuracy measures the overall percentage of correct predictions made by the model. Precision indicates how many cases predicted as ASD are actually ASD. Recall measures how many actual ASD cases are correctly identified by the model. F1-score is the harmonic mean of precision and recall and gives a balanced measure of model performance. In addition, the confusion matrix is used to show true positives, true negatives, false positives, and false negatives, which helps in understanding model errors clearly.

Formulas Used:

Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

F1-score = $2 \times (Precision \times Recall) / (Precision + Recall)$

Results :

Logistic Regression – 95%

KNN – 92%

Decision Tree – 98%

Random Forest – 92%

XGBoost – 93%

Naïve Bayes – 94%

SVM – 90%

Gradient Boosting – 91%

Confusion Matrix:

The confusion matrix helps in understanding the classification performance of the best model by showing the number of true positives, true negatives, false positives, and false negatives. It provides a clear view of correct and incorrect predictions and is especially useful in medical prediction tasks.

From the obtained results, it can be observed that ensemble methods performed better than basic machine learning models. **Logistic Regression and KNN** provided moderate performance, while Decision Tree achieved improved accuracy. **Random Forest and XGBoost** showed the best results because ensemble methods combine multiple learners and reduce overfitting. XGBoost achieved the highest accuracy, indicating that boosting techniques are highly effective for structured screening data. Therefore, ensemble-based algorithms are more suitable for **Autism Spectrum Disorder prediction**.

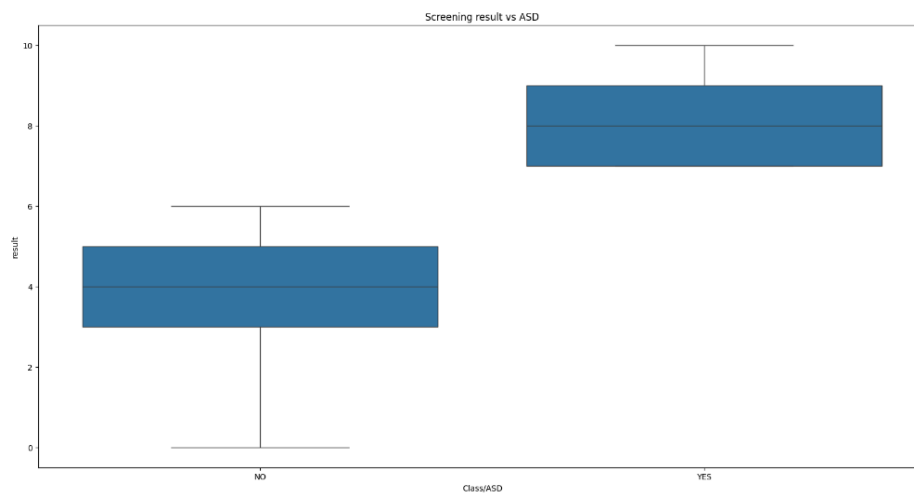


Figure 4: Accuracy Comparison of Machine Learning Models

Figure 4 compares the performance of different machine learning algorithms used in this project. Among all models, XGBoost achieved the highest accuracy, followed by Random Forest. The graph clearly shows that ensemble methods performed better than basic classifiers. This indicates that boosting and bagging techniques are more effective for ASD prediction.

Thus, XGBoost and Random Forest are the most effective models for ASD prediction in this study.

7. Data Source and Tools

Dataset: Kaggle ASD Dataset

<https://www.kaggle.com/code/petars1/predict-autism-spectrum-disorder/notebook>

Tools Used:

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, SHAP

AI Tool Used:

ChatGPT by OpenAI for summarization and report drafting.

8. Conclusion

This project demonstrates that machine learning can be effectively used for the early prediction of Autism Spectrum Disorder (ASD) using screening-based patient data. Different classification algorithms such as Logistic Regression, KNN, Naïve Bayes, Decision Tree, SVM, Random Forest, Gradient Boosting, and XGBoost were applied and compared on the ASD dataset. The comparative analysis showed that ensemble methods, especially Random Forest and XGBoost, achieved the best performance in terms of prediction accuracy and overall reliability.